Downscaling the Twentieth Century Reanalysis over India

WRF and RegCM5 on the 1924 Kerala flood and the 26 July 2005 Mumbai cloudburst, scored against criteria fixed in advance

Likith Muni Narakala
[affiliation]

White paper, October 2026

Abstract

Before 1940, atmospheric reanalyses rest on surface observations alone. The NOAA–CIRES–DOE Twentieth Century Reanalysis version 3 (20CRv3) assimilates only surface pressure. We test whether two regional climate models, WRF 4.5 and RegCM5, driven by 20CRv3 at 12 and 3 km grid spacing, can reconstruct extreme west-coast monsoon rainfall over India. Two events are simulated: the Kerala flood of July 1924 and the Mumbai cloudburst of 26 July 2005, the second also driven by ERA5 to separate driver error from model error. Within each event all runs are scored on the same India Meteorological Department (IMD) 0.25° cells, against five criteria committed before any of the nine runs was started. For 1924, downscaling turns the driver's weak, smooth rainfall (bias −80%, pattern correlation 0.51) into an orographic band along the Western Ghats (correlation 0.80–0.83 and bias −39% at 12 km) and adds skill in all four runs, but the modelled rain peaks three days after the observed peak. For 2005, no model, resolution or driver reproduces the cloudburst: the largest 24 h total of any run within 50 km of Santacruz is 163 mm, against 944 mm observed. The 3 km grids did not consistently improve on 12 km. The spatial pattern of rainfall held in place by the terrain can be reconstructed from 20CRv3, although too dry and too late; an organised convective extreme over the coastal plain was not reproduced in this configuration. We also document an out-of-bounds table read in the WRF Thompson microphysics that crashed every WRF run, and its fix. Code, namelists, scores and figures are public.

1 Introduction

Extreme rainfall events from before the satellite era are poorly known beyond the station records that captured them. Reanalyses that assimilate upper-air observations begin in the 1940s; ERA5 now reaches back to 1940. Earlier years are covered only by reanalyses that assimilate surface observations alone: the Twentieth Century Reanalysis (Compo et al., 2011), ERA-20C (Poli et al., 2016) and the coupled CERA-20C (Laloyaux et al., 2018). We use the most recent of these, version 3 of the Twentieth Century Reanalysis (20CRv3; Slivinski et al., 2019), which assimilates only surface and sea-level pressure observations, with prescribed sea-surface temperature and sea ice, in an 80-member ensemble. Its circulation is therefore anchored by observations, but its moisture and rainfall are products of the forecast model; its performance is evaluated by Slivinski et al. (2021). On the 1° grid on which it is distributed, the orography that shapes monsoon rainfall on the west coast of India is barely resolved.

Dynamical downscaling can add that missing detail. WRF has been driven by the Twentieth Century Reanalysis for three Vb cyclones over the Alps, of 2005, 1910 and 1876 (Stucki et al., 2020), and by 20CRv3, with and without assimilation of early station observations, for a cold-air outbreak in the 1816 "year without a summer" (Stucki et al., 2024); 20CR has also been downscaled over China for present-day climate means (Kong and Bi, 2015). To our knowledge no study has downscaled 20CRv3 for a South Asian monsoon extreme, and none of the reanalysis

downscaling studies above compares two regional models on the same verification cells or fixes its success criteria before the simulations.

This white paper reports such an experiment. Two regional models, WRF (Skamarock et al., 2019) and RegCM5 (Giorgi et al., 2023), are run at 12 km (parameterised convection) and 3 km (convection-permitting) for one pre-1950 event, before ERA5 begins, and for one recent event, when the same models can also be driven by ERA5 (Hersbach et al., 2020). The recent event separates what the models can do from what the historical driver allows. Five statements (H1–H5, Section 4) were committed to the repository before any of the nine runs was started, and all outcomes are reported, failures included.

2 Events

E1: Kerala, July 1924. The "great flood of '99" (Malayalam year 1099) followed heavy monsoon rain over the Western Ghats of central Kerala; the Periyar basin counts it as one of only two severe floods in its recent history (Sudheer et al., 2019), and the 2018 floods were described as Kerala's worst since 1924 (Hunt and Menon, 2020). A Central Water Commission study describes the storm of 16–18 July 1924 over the High Ranges (Devikulam and Munnar), with a three-day Kerala-average rainfall of 443 mm, more than the 414 mm of 15–17 August 2018 (Central Water Commission, 2018). ERA5 does not reach back to this event.

E2: Mumbai, 26 July 2005. The Santacruz observatory recorded 944 mm in the 24 h ending 08:30 IST on 27 July 2005, while Colaba in south Mumbai recorded 73 mm (Kumar et al., 2008). In WRF simulations of the event, the Western Ghats and the land–sea contrast together enhanced the rain by more than half, and an offshore mesoscale vortex depended on latent heating (Kumar et al., 2008). This event is driven by both 20CRv3 and ERA5.

Both events are west-coast monsoon rainfall. Between them the era changes, and with it the density of observations behind 20CRv3; so do the location and, as shown below, the mechanism of the heaviest rain. Rainfall was chosen because IMD gridded daily rainfall exists from 1901 (Pai et al., 2014), so both eras can be verified on the same grid.

3 Models and set-up

Both models share one 12 km domain (271×241 points, Mercator projection, 1.8–27.5°N, 60.9–91.1°E) and use a 3 km domain of 241×241 points centred on each event's scoring box (Fig. 1). Each model runs in its reference configuration (Table 1), so the comparison is between model systems rather than single parameterisations. Neither model is nudged in the interior: the drivers act through the initial state, the lateral boundaries and the sea-surface temperature. E1 runs from 00 UTC 12 July to 00 UTC 21 July 1924 (216 h) and E2 from 00 UTC 24 July to 00 UTC 30 July 2005 (144 h), each with 27 h of spin-up before the first scored day.

20CRv3 is used as its ensemble mean (1°, 28 pressure levels, 3-hourly) and ERA5 at 0.25° on 37 pressure levels, sub-sampled to 3-hourly. Both drivers pass through the same code path in each model: Python writes WPS intermediate files for WRF and files in the ERA5 layout of the RegCM5 pre-processor (which has no 20CRv3 reader). The nine runs are WRF (12 and 3 km in one run) and RegCM5 at 12 and 3 km for each of E1 with 20CRv3, E2 with 20CRv3 and E2 with ERA5; each driver's own rainfall is scored as the no-downscaling baseline.

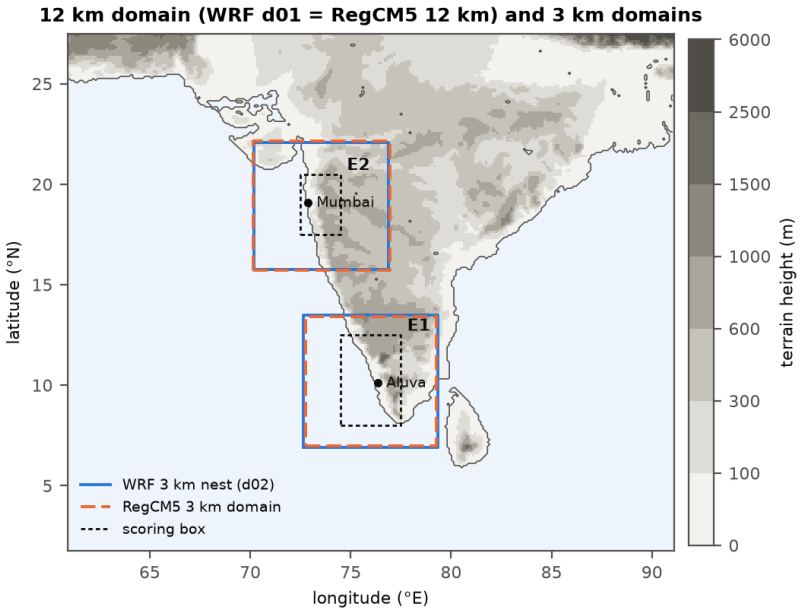


Figure 1. The shared 12 km domain (WRF d01 and the RegCM5 12 km grid) over terrain, the 3 km domains of each model for each event (WRF blue, RegCM5 dashed orange) and the scoring boxes (black, finely dashed).

Table 1. Model configurations.

	WRF-ARW 4.5	RegCM5 (MOLOCH non-hydrostatic core)
Nesting	d01 12 km and d02 3 km in one run, one-way	3 km run nested on the 12 km run's output
Levels, top	50, 50 hPa	30 (12 km), 50 (3 km)
Time step	60 s / 15 s	90 s / 30 s
Microphysics	Thompson (Thompson et al., 2008), with the bounds fix of Section 7.1	Nogherotto–Tompkins (Nogherotto et al., 2016)
Convection	Grell–Freitas (Grell and Freitas, 2014) at 12 km, none at 3 km	Tiedtke (Tiedtke, 1989) at 12 km, none at 3 km
PBL	YSU (Hong et al., 2006)	Holtslag
Land surface	Noah, soil from the driver	BATS, RegCM5 default initial soil
Radiation	RRTMG (Iacono et al., 2008), no aerosol	CCSM, no aerosol
SST	held at the initial value	updated from the driver every 3 h
Lateral boundary	5-point relaxation zone	exponential relaxation, 20 (12 km) / 31 (3 km) points

4 Verification and criteria fixed in advance

All rainfall is compared on the IMD 0.25° daily grid (Pai et al., 2014), land cells only. Model and driver rainfall are remapped onto it with an area-weighted (first-order conservative) remap, so every run is scored on identical cells inside each event's scoring box (Fig. 1). Days are 24 h windows ending at 03 UTC (08:30 IST, the IMD observation time): seven days for E1 (13–20 July 1924) and four for E2 (25–29 July 2005). The date convention of the IMD files was not assumed but read from the data: the 2005 box maximum (732.5 mm) falls on the file's 27 July label, the morning on which the Santacruz record was reported, so a label marks the end of its 24 h window.

The metrics are the bias of the box-mean event total; the area-weighted pattern correlation of event totals (r); the ratio of the largest daily cell value to the observed one (peak ratio); the distance between the event-total maxima; the SAL structure, amplitude and location components (Wernli et al., 2008); and the fractions skill score (FSS; Roberts and Lean, 2008) pooled over the scored days at the IMD heavy, very heavy and extremely heavy rainfall thresholds (64.5, 115.6 and 204.5 mm day^{-1}), here with a 5-point (1.25°) neighbourhood unless stated. FSS is "useful" at or above $0.5 + f_0/2$, with f_0 the observed fraction of cells above the threshold (Roberts and Lean, 2008). Added value (AV) is a run's score minus the score of its own driver's rainfall on the same cells.

The five criteria below were committed to the repository on 15 October 2026 (UTC; commit `260f8c9`, `docs/metrics.md`), before any of the nine runs was started, and the file has not been changed since.

H1 The models can produce E2 when the driver is good: with ERA5 at 3 km, $\mathrm{FSS}_{115.6}$ at or above useful and $|\mathrm{bias}| \leq 30\%$ for at least one model.

H2 The historical driver costs little: for each model and resolution in E2, the $\mathrm{FSS}_{115.6}$ with ERA5 exceeds that with 20CRv3 by at most 0.15, and $|\mathrm{bias}| \leq 30\%$ with 20CRv3.

H3 Downscaling adds value: AV in $\mathrm{FSS}_{64.5}$ is positive for every run.

H4 3 km beats 12 km on extremes: for each model and case, a peak ratio closer to 1 and a higher $\mathrm{FSS}_{115.6}$.

H5 The pre-1950 event is reproduced: in E1 at 3 km, $\mathrm{FSS}_{64.5}$ (9-point neighbourhood) at or above useful and $|\mathrm{bias}| \leq 40\%$ for at least one model.

5 Results

5.1 E1: the 1924 rain pattern rebuilt from surface pressure

20CRv3 holds the large-scale ingredients of an active monsoon spell on the wettest day: a moist column over the Arabian Sea, strong 850 hPa westerlies onto the Kerala coast and a monsoon low near the head of the Bay of Bengal (Fig. 2). Its own rainfall, however, is a smooth field with no maximum on the Ghats: bias −80%, pattern correlation 0.51, peak ratio 0.17, and no cell in the box above about 60 mm day^{-1} on any day (Figs. 3 and 4, Table 2).

Every downscaled run puts a band of heavy rain along the Western Ghats. At 12 km both models reach a pattern correlation of 0.80–0.83 with a bias of −39%; WRF places its event-total maximum 27 km from the observed one. Peak ratios rise from 0.17 to 0.52–0.62, and the added value in $\mathrm{FSS}_{64.5}$ is +0.58 to +0.71 (H3 passes for all four runs). The FSS at 204.5 mm day^{-1} is zero for every run.

The main shortcoming is timing (Fig. 4). The observed box mean rises to about 104 mm on IMD day 17 July, which matches the storm of 16–18 July, and falls afterwards. Every downscaled run, and 20CRv3's own rainfall, has its wettest day on 20 July, the last scored day and three days

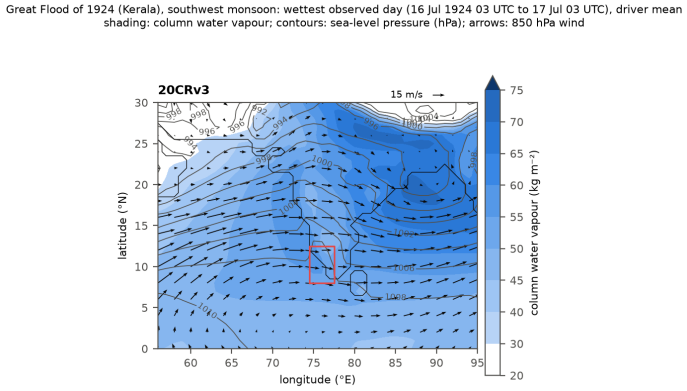


Figure 2. 20CRv3 ensemble mean on the wettest observed day of E1 (03 UTC 16 to 03 UTC 17 July 1924): column water vapour (shading), sea-level pressure (contours, hPa) and 850 hPa wind (arrows). The red box is the scoring box; the black line is 20CRv3's own 1° coastline.

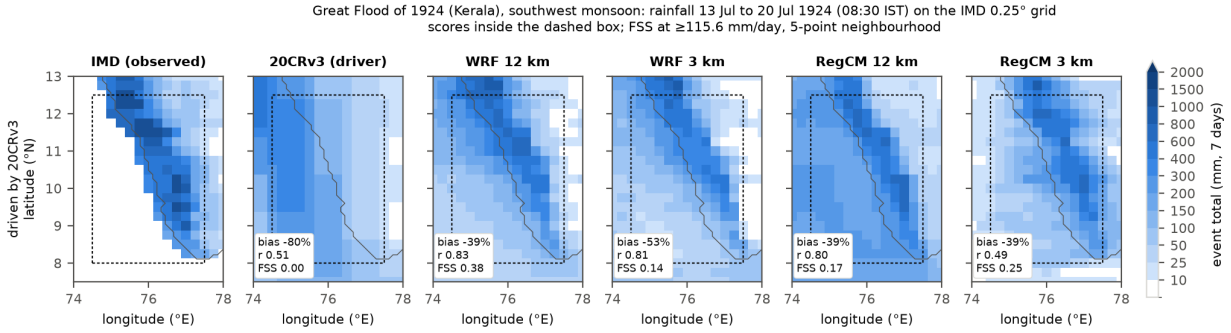


Figure 3. E1 rainfall, 13–20 July 1924, on the IMD grid: observed, the 20CRv3 rainfall, and each downscaled run. Each panel gives its bias, pattern correlation and FSS at 115.6 mm day^{-1} inside the scoring box.

Table 2. E1 scores, generated from the scoring output in `results/`. FSS with a 5-point neighbourhood; S, A, L are the SAL components.

Run	Driver	Bias (%)	r	Peak	Max. dist. (km)	S	A	L	$FSS_{115.6}$	$FSS_{204.5}$
perfect		0	1	1	0	0	0	0	1	1
driver alone	20CRv3	−80	0.51	0.17	83	−0.13	−1.32	0.08	0.00	0.00
WRF 12 km	20CRv3	−39	0.83	0.62	27	+0.05	−0.48	0.04	0.38	0.00
WRF 3 km	20CRv3	−53	0.81	0.56	54	+0.03	−0.71	0.03	0.14	0.00
RegCM5 12 km	20CRv3	−39	0.80	0.52	194	+0.02	−0.49	0.04	0.17	0.00
RegCM5 3 km	20CRv3	−39	0.49	0.62	315	+0.17	−0.49	0.06	0.25	0.00

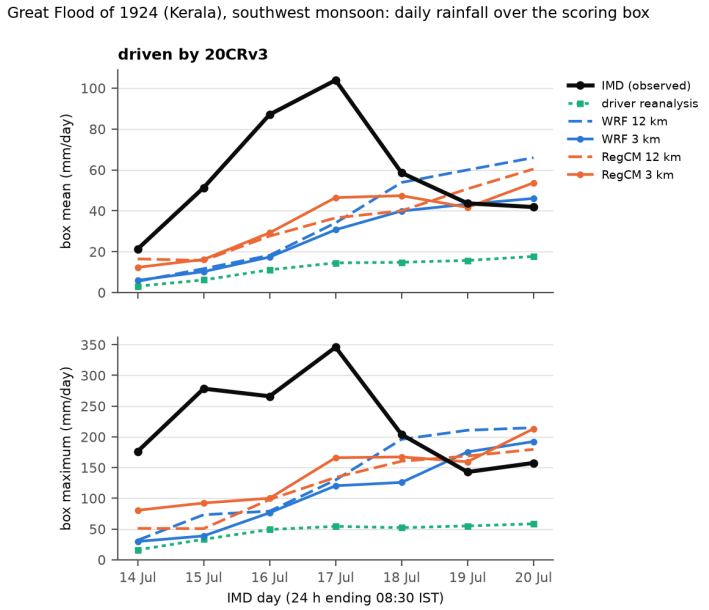


Figure 4. E1 day by day over the scoring box: area mean (top) and largest cell value (bottom). IMD days end at 08:30 IST on the date shown.

after the observed peak, when the observed box mean is below half of its peak; their largest daily cell values reach 0.52–0.62 of the observed one. Because the driver shows the same late build-up, the lag most likely enters through the boundaries, from the timing of the monsoon surge in 20CRv3, rather than from either regional model.

At 3 km, RegCM5 meets H5 by a narrow margin ($FSS_{64.5}$ with a 9-point neighbourhood 0.70 against a useful level of 0.68; bias −39% against a limit of 40%); WRF at 3 km does not (0.57; −53%).

5.2 E2: the cloudburst that no run produced

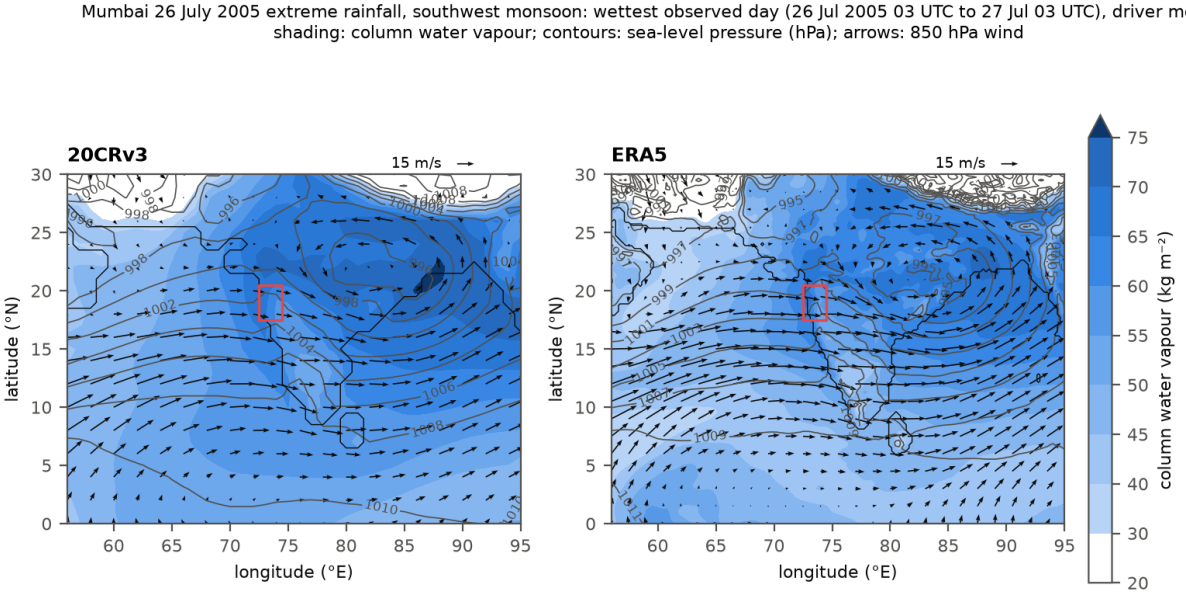


Figure 5. As Fig. 2 for E2 (03 UTC 26 to 03 UTC 27 July 2005), 20CRv3 (left) and ERA5 (right).

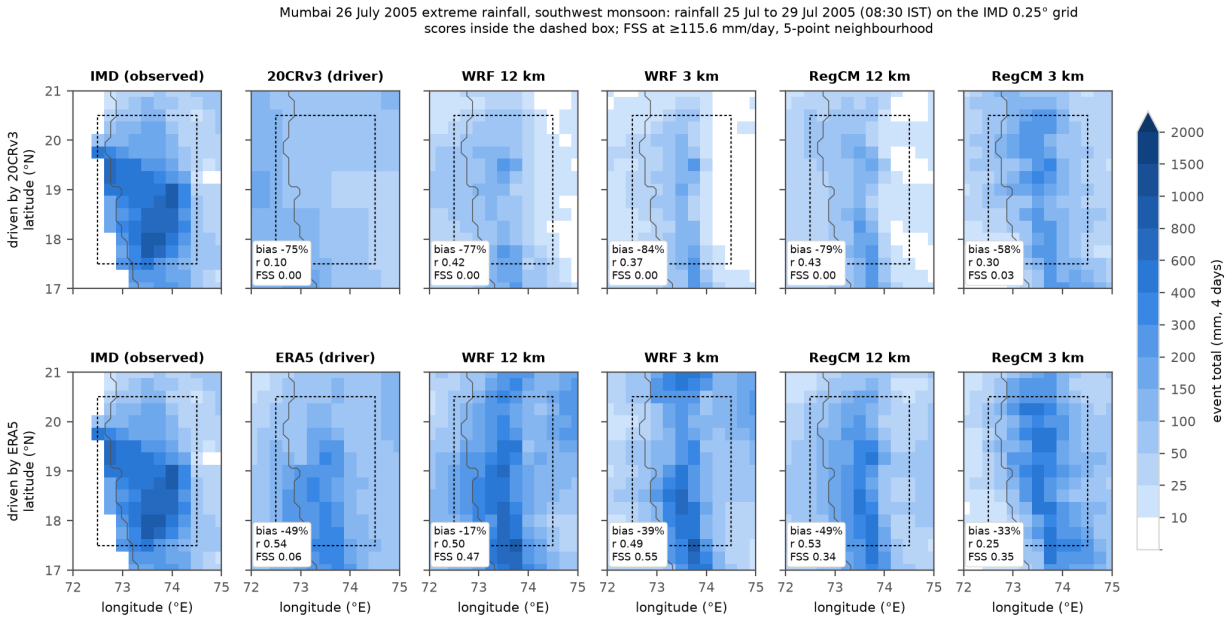


Figure 6. As Fig. 3 for E2, 25–29 July 2005, driven by 20CRv3 (top) and ERA5 (bottom).

Both drivers place a monsoon low over central and eastern India with strong westerlies onto the Konkan coast on 26 July (Fig. 5); the large-scale setting for heavy rain is present in both. The observed four-day rainfall covers the coastal plain and the Ghats between about 17.5 and 19.5°N, Mumbai included (Fig. 6). The ERA5-driven runs concentrate their rain in a narrow north–south band on the Ghats crest and leave the coast with a fraction of the observed amount. They still give the best E2 scores: WRF at 12 km has a bias of −17%, and WRF at 3 km an $FSS_{115.6}$ of 0.55, just below the useful level of 0.60 (Table 3). Driven by 20CRv3, all four runs are 58–84% too dry.

Table 3. As Table 2 for E2.

Run	Driver	Bias (%)	*r*	Peak	Max. dist. (km)	S	A	L	$FSS_{115.6}$	$FSS_{204.5}$
perfect		0	1	1	0	0	0	0	1	1
driver alone	20CRv3	−75	0.10	0.06	77	+0.54	−1.20	0.09	0.00	0.00
WRF 12 km	20CRv3	−77	0.42	0.10	28	−0.11	−1.26	0.05	0.00	0.00
WRF 3 km	20CRv3	−84	0.37	0.10	62	−0.55	−1.45	0.05	0.00	0.00
RegCM5 12 km	20CRv3	−79	0.43	0.10	62	−0.24	−1.29	0.05	0.00	0.00
RegCM5 3 km	20CRv3	−58	0.30	0.17	139	+0.12	−0.81	0.11	0.03	0.00
driver alone	ERA5	−49	0.54	0.18	56	+0.25	−0.64	0.02	0.06	0.00
WRF 12 km	ERA5	−17	0.50	0.46	62	−0.05	−0.19	0.04	0.47	0.20
WRF 3 km	ERA5	−39	0.49	0.54	62	−0.38	−0.48	0.04	0.55	0.31
RegCM5 12 km	ERA5	−49	0.54	0.23	62	+0.02	−0.65	0.04	0.34	0.00
RegCM5 3 km	ERA5	−33	0.26	0.41	111	+0.08	−0.39	0.10	0.35	0.04

Day by day (Fig. 7), the ERA5-driven WRF runs come close to the observations on the day before the event (WRF 3 km box maximum about 400 mm against about 415 mm). On the cloudburst day, IMD day 27 July, the observed box mean jumps to about 137 mm and the box maximum to 732.5 mm; every ERA5-driven run goes down instead, and the 20CRv3-driven runs stay flat. At Santacruz itself (Fig. 8), no run exceeds 100 mm in the 24 h in which 944 mm was measured. The largest 24 h total of any run within 50 km is 163 mm (RegCM5 3 km, 20CRv3); with ERA5 it is 91 mm (WRF 3 km).

H1 therefore fails: the best ERA5-driven 3 km run (WRF) reaches $FSS_{115.6} = 0.55$ against 0.60, with a bias of −39% against a limit of 30%. H2 fails for every model and resolution: replacing ERA5 by 20CRv3 lowers $FSS_{115.6}$ by 0.31–0.55 and leaves biases of −58 to −84%. For H3, the

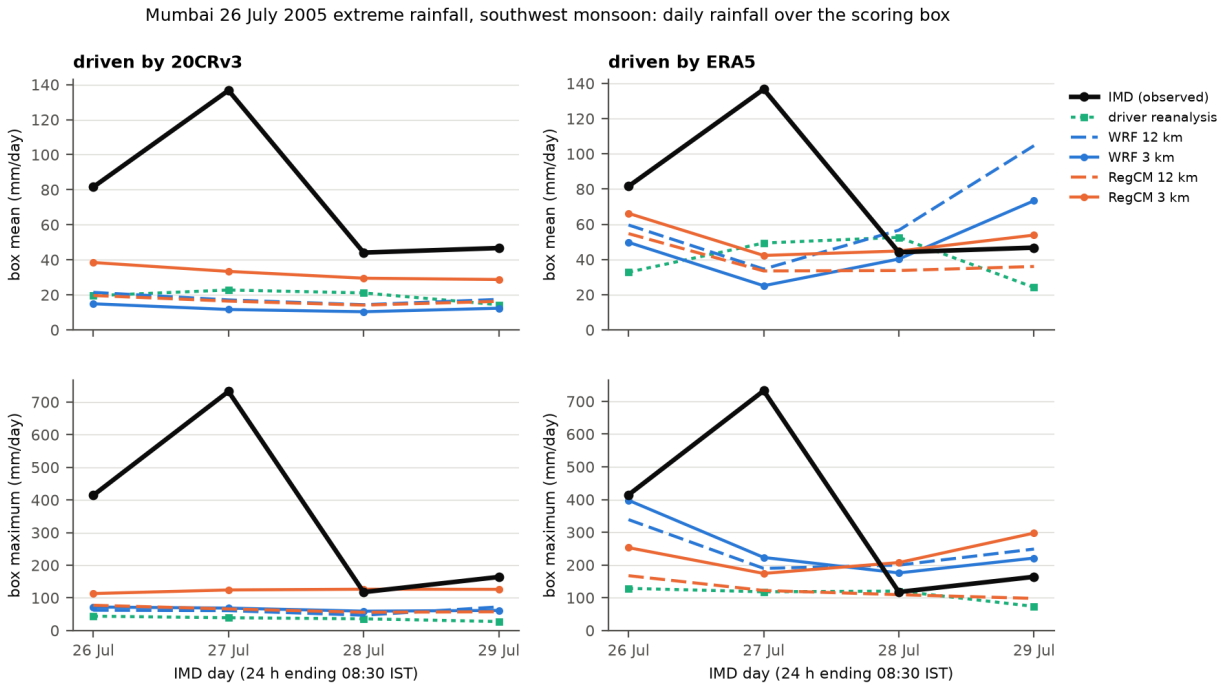


Figure 7. As Fig. 4 for E2, driven by 20CRv3 (left) and ERA5 (right).

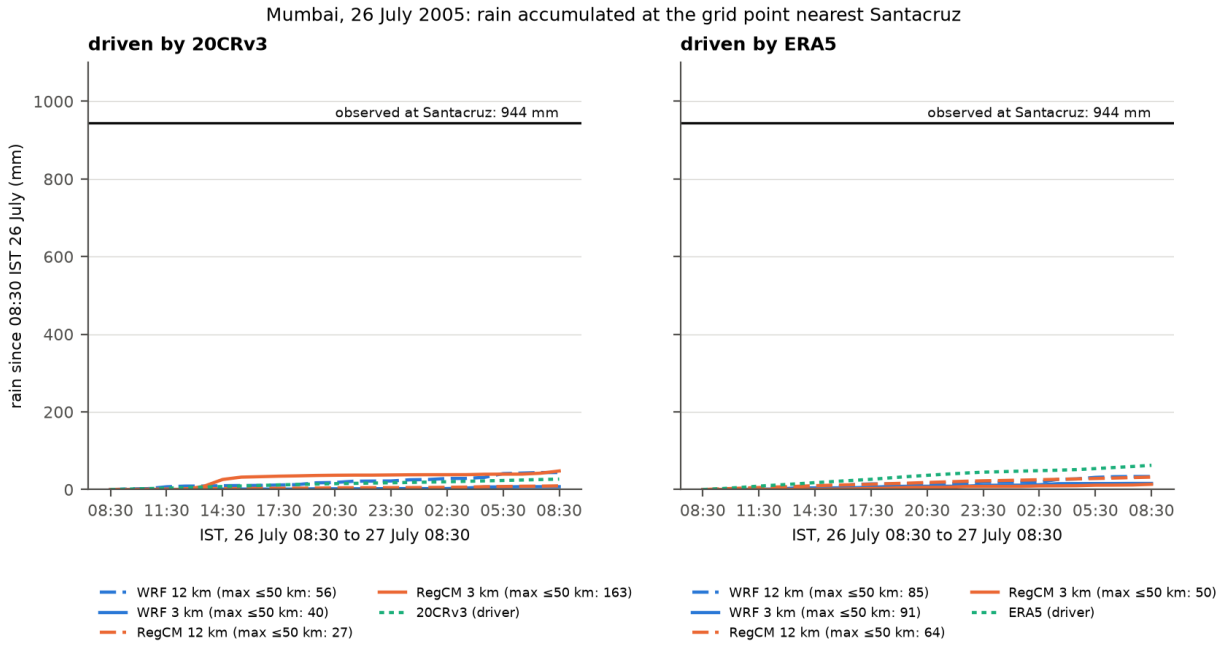


Figure 8. Rain accumulated at the model grid point nearest the Santacruz observatory from 08:30 IST 26 July to 08:30 IST 27 July 2005; the legend gives each run's largest 24 h total within 50 km of Santacruz. The black line is the 944 mm observed.

20CRv3-driven runs add value over their driver, but by only +0.01 to +0.05 in three of the four (RegCM5 3 km: +0.29), while all four ERA5-driven runs fall below ERA5's own $FSS_{64.5}$ of 0.70 (−0.02 to −0.09). At the higher 115.6 mm threshold, where ERA5 alone scores 0.06, the ERA5-driven runs score 0.34–0.55.

5.3 Does 3 km help?

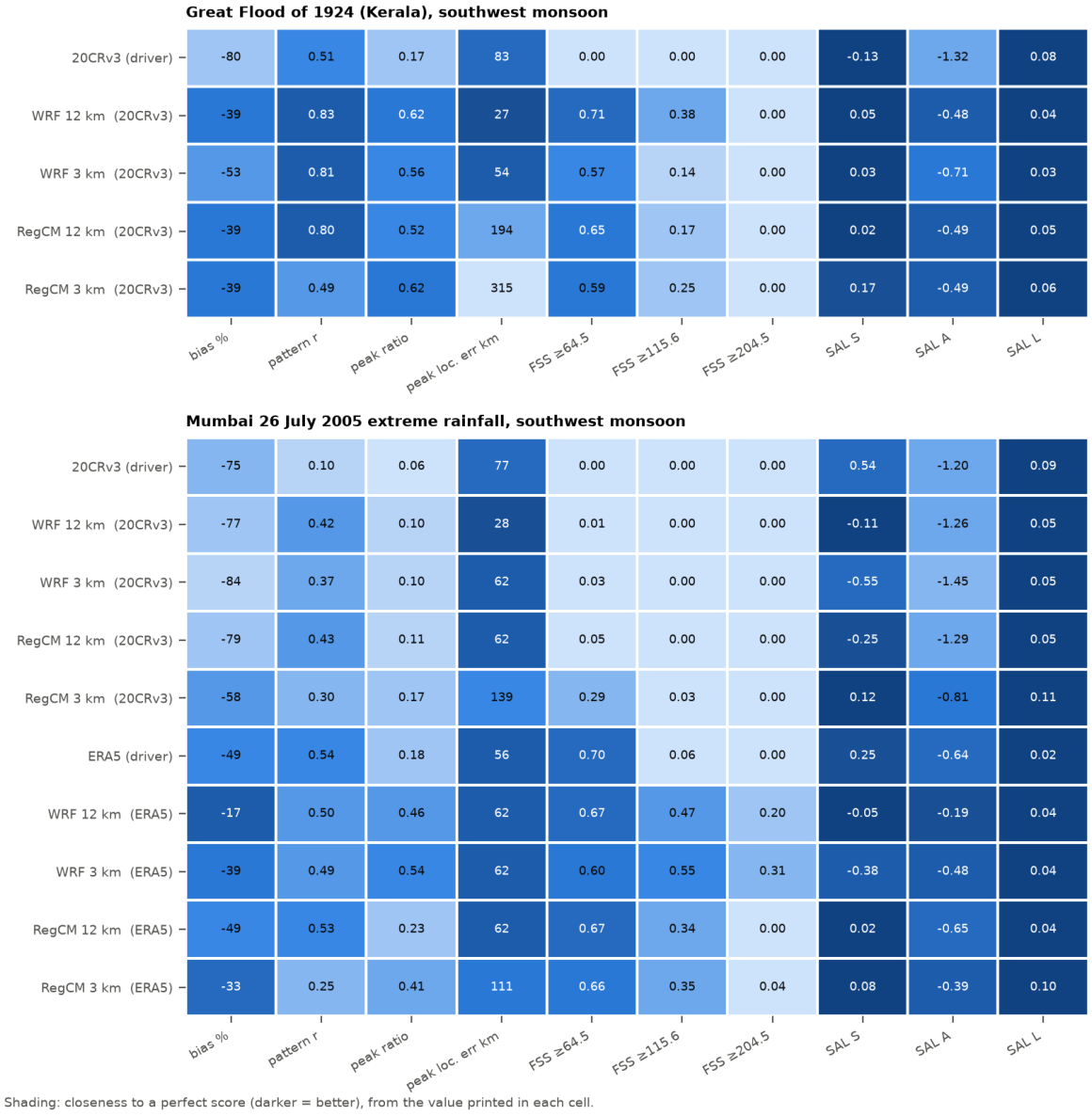


Figure 9. All headline scores. Shading shows closeness to a perfect score (darker is better), computed from the value printed in each cell.

H4 passes in four of six cases (Fig. 9). WRF at 3 km, with no cumulus scheme, is drier than WRF at 12 km with Grell–Freitas in every case: E1 −39 to −53%, E2 with 20CRv3 −77 to −84%, E2 with ERA5 −17 to −39%. With ERA5 it still sharpens the extremes (peak ratio 0.46 to 0.54, $FSS_{115.6}$ 0.47 to 0.55), but with 20CRv3 in 1924 it loses them ($FSS_{115.6}$ 0.38 to 0.14). RegCM5 at 3 km is as wet as or wetter than at 12 km and raises the peak ratio in all three cases (E1 0.52 to 0.62; E2 20CRv3 0.11 to 0.17; E2 ERA5 0.23 to 0.41), but its pattern correlation drops (E1 0.80 to 0.49; E2 ERA5 0.53 to 0.25) and its event-total maximum lies 111–315 km from the observed one (native 3 km fields in Fig. 10).

6 Discussion

Terrain-held rain versus organised convection. The contrast between the two events is the main result. In 1924, the heaviest rain lay along the Ghats, where the lifting is set by the terrain; a regional model given roughly the right onshore flow puts rain in the right place, even from a 1° reanalysis that assimilates only pressure. In 2005, the extreme came from organised convection over the coastal plain, which depends on mesoscale details that neither driver supplies directly and that the models did not generate. The ERA5-driven runs placed their rain on the Ghats crest, as for 1924, and missed the coastal maximum.

What the driver sets. The late build-up in 1924 is shared by 20CRv3's own rainfall and every downscaled run, so the regional models inherited it. The 2005 synoptic setting is similar in both drivers (Fig. 5), yet the 20CRv3-driven runs are far drier than the ERA5-driven ones (H2). The difference most likely lies in what the regional models receive from the driver (initial state, lateral boundaries or sea-surface temperature), for example the strength and moisture of the low-level flow; we have not diagnosed it. Ensemble members of 20CRv3, rather than its mean, would show how much of either error is chance.

Added value depends on the driver and the threshold. Over a 1° driver without orographic rain, every run adds value. Over ERA5, which already scores FSS 0.70 at 64.5 mm, downscaling lowers the FSS at that threshold by 0.02–0.09 but adds +0.28 to +0.49 at 115.6 mm, consistent with the view that a regional model can add value mainly at scales the driver does not resolve (Di Luca et al., 2012).

Resolution and physics. Convection-permitting grids are expected to improve heavy, short-duration rainfall (Prein et al., 2015); with two cases and daily totals we see mixed results, and WRF without a cumulus scheme rained less than WRF with one in every case. The 2005 event is known to be sensitive to the physics: with WRF from the same start time, Kumar et al. (2008), with the Grell–Dévényi cumulus scheme (Grell and Dévényi, 2002) on their 33 and 11 km domains, WSM6 microphysics and an explicit 3.6 km nest, captured the Santacruz rainfall; with Kain–Fritsch or Betts–Miller–Janjić on the outer domains the rain was underestimated by more than half. Our WRF configuration (Thompson and Grell–Freitas) is one that misses it; whether the physics alone explains the miss was not tested here.

7 Technical findings

7.1 An out-of-bounds read in WRF Thompson microphysics

All three WRF runs first stopped with segmentation faults in the Thompson microphysics (`mp_physics = 8`): E1 after 117.6 simulated hours, E2 with 20CRv3 after 74 h and E2 with ERA5 after 8.3 h, with no CFL warnings and ample memory. All three faulted at the same machine instruction. In WRF 4.5, when hail-aware graupel is off, the rain–graupel collection tables are allocated with a graupel-density dimension of extent 1, but the eight lookups into them use a graupel-density index of 5. Each lookup therefore reads the wrong table entry, and in the heaviest rain with graupel it reads beyond the end of the table. A one-column test of WRF 4.5's own module with Fortran bounds checking stops at one of these lookups. The same lookups are unchanged in WRF 4.5.1 to 4.7.1. Our fix clamps the index to the table extent (`MIN(idx_bg(k), dimNRHG)`), which leaves the hail-aware scheme (`mp_physics = 38`) unchanged, and adds the two Thompson changes of WRF 4.5.1. With it, all three runs completed from the initial time without a single stop. The patch and the one-column reproducer are in the repository.

7.2 Reading the observation date convention from the data

The first scoring run used an assumed IMD date convention that turned out to be wrong by one day, comparing every model day with the following observed day. A check that reads the convention from the 2005 record (Section 4) caught it; the first scorecard was discarded, and the scoring code now refuses to run when the check fails.

7.3 Cost

The nine runs used 1,891 core-hours, two-thirds of them for WRF, against a planning estimate of 741–1,482, on a shared node with more processes than cores. WRF took 28–30 h of wall time per run on 12–20 cores; RegCM5 took 1.4–3.2 h at 12 km and 5.9–13.2 h at 3 km on 20 cores.

8 Limitations

Two events are case studies, and the criteria are pass/fail statements per case, without significance testing. Each event is one deterministic run from the 20CRv3 ensemble mean. H5 passes by 0.02 in FSS and 0.6 percentage points of bias. Other century-long reanalyses (ERA-20C, CERA-20C) were not tried as drivers. The 1924 IMD grid rests on fewer gauges than the 2005 grid. Present-day land use is used for 1924, and the two models initialise soil moisture differently. The Santacruz figure is a station total, compared with grid-point and 50 km values.

9 Outlook

- **Ensembles of the driver.** Downscaling a set of 20CRv3 members would give a distribution of rainfall against which an observed extreme can be judged, and separate driver uncertainty from chance in convection.
- **Physics as part of the ensemble.** Starting with the Grell–Dévényi and WSM6 combination of Kumar et al. (2008) in this set-up.
- **Bias correction** of the downscaled pre-1950 rainfall (-39% in three of four E1 runs), tested on further IMD-era monsoon extremes before 1950.

Code and data availability

Code, configuration, namelists, scores and figures: [GitHub URL]; archived version: [Zenodo DOI]. 20CRv3 is available from NOAA PSL, ERA5 from the Copernicus Climate Change Service, and the IMD gridded rainfall from the India Meteorological Department.

Acknowledgements

Support for the Twentieth Century Reanalysis Project version 3 dataset is provided by the U.S. Department of Energy, Office of Science Biological and Environmental Research (BER), by the National Oceanic and Atmospheric Administration Climate Program Office, and by the NOAA Physical Sciences Laboratory. ERA5 is produced by the Copernicus Climate Change Service at ECMWF. Rainfall observations are from the India Meteorological Department. WRF is developed by NCAR and RegCM5 by ICTP.

References

Central Water Commission. Kerala floods of August 2018. Study report, Hydrology (S) Directorate, Central Water Commission, Government of India, Sept. 2018.

G. P. Compo, J. S. Whitaker, P. D. Sardeshmukh, et al. The Twentieth Century Reanalysis Project. *Quarterly Journal of the Royal Meteorological Society*, 137(654):1–28, 2011. doi: 10.1002/qj.776.

A. Di Luca, R. de Elía, and R. Laprise. Potential for added value in precipitation simulated by high-resolution nested Regional Climate Models and observations. *Climate Dynamics*, 38(5-6): 1229–1247, 2012. doi: 10.1007/s00382-011-1068-3.

F. Giorgi, E. Coppola, G. Giuliani, J. M. Ciarlò, E. Pichelli, R. Nogherotto, F. Raffaele, P. Malguzzi, S. Davolio, P. Stocchi, and O. Drofa. The fifth generation regional climate modeling system, RegCM5: Description and illustrative examples at parameterized convection and convection-permitting resolutions. *Journal of Geophysical Research: Atmospheres*, 128(6):e2022JD038199, 2023. doi: 10.1029/2022JD038199.

G. A. Grell and D. Dévényi. A generalized approach to parameterizing convection combining ensemble and data assimilation techniques. *Geophysical Research Letters*, 29(14):1693, 2002. doi: 10.1029/2002GL015311.

G. A. Grell and S. R. Freitas. A scale and aerosol aware stochastic convective parameterization for weather and air quality modeling. *Atmospheric Chemistry and Physics*, 14(10):5233–5250, 2014. doi: 10.5194/acp-14-5233-2014.

H. Hersbach, B. Bell, P. Berrisford, et al. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049, 2020. doi: 10.1002/qj.3803.

S.-Y. Hong, Y. Noh, and J. Dudhia. A new vertical diffusion package with an explicit treatment of entrainment processes. *Monthly Weather Review*, 134(9):2318–2341, 2006. doi: 10.1175/MWR3199.1.

K. M. R. Hunt and A. Menon. The 2018 Kerala floods: a climate change perspective. *Climate Dynamics*, 54(3-4):2433–2446, 2020. doi: 10.1007/s00382-020-05123-7.

M. J. Iacono, J. S. Delamere, E. J. Mlawer, M. W. Shephard, S. A. Clough, and W. D. Collins. Radiative forcing by long-lived greenhouse gases: Calculations with the AER radiative transfer models. *Journal of Geophysical Research*, 113(D13):D13103, 2008. doi: 10.1029/2008JD009944.

X.-H. Kong and X.-Q. Bi. Dynamical downscaling of the Twentieth Century Reanalysis for China: Climatic means during 1981–2010. *Atmospheric and Oceanic Science Letters*, 8(3):166–173, 2015. doi: 10.1080/16742834.2015.11447255.

A. Kumar, J. Dudhia, R. Rotunno, D. Niyogi, and U. C. Mohanty. Analysis of the 26 July 2005 heavy rain event over Mumbai, India using the Weather Research and Forecasting (WRF) model. *Quarterly Journal of the Royal Meteorological Society*, 134(636):1897–1910, 2008. doi: 10.1002/qj.325.

P. Laloyaux, E. de Boisseson, M. Balmaseda, et al. CERA-20C: A coupled reanalysis of the twentieth century. *Journal of Advances in Modeling Earth Systems*, 10(5):1172–1195, 2018. doi: 10.1029/2018MS001273.

R. Nogherotto, A. M. Tompkins, G. Giuliani, E. Coppola, and F. Giorgi. Numerical framework and performance of the new multiple-phase cloud microphysics scheme in RegCM4.5: precipitation, cloud microphysics, and cloud radiative effects. *Geoscientific Model Development*, 9(7):2533–2547, 2016. doi: 10.5194/gmd-9-2533-2016.

D. S. Pai, L. Sridhar, M. Rajeevan, O. P. Sreejith, N. S. Satbhai, and B. Mukhopadhyay. Development of a new high spatial resolution ($0.25^\circ \times 0.25^\circ$) long period (1901–2010) daily gridded rainfall data set over India and its comparison with existing data sets over the region. *MAUSAM*, 65(1): 1–18, 2014. doi: 10.54302/mausam.v65i1.851.

P. Poli, H. Hersbach, D. P. Dee, et al. ERA-20C: An atmospheric reanalysis of the twentieth century. *Journal of Climate*, 29(11):4083–4097, 2016. doi: 10.1175/JCLI-D-15-0556.1.

A. F. Prein, W. Langhans, G. Fosser, et al. A review on regional convection-permitting climate modeling: Demonstrations, prospects, and challenges. *Reviews of Geophysics*, 53(2):323–361, 2015. doi: 10.1002/2014RG000475.

N. M. Roberts and H. W. Lean. Scale-selective verification of rainfall accumulations from high-resolution forecasts of convective events. *Monthly Weather Review*, 136(1):78–97, 2008. doi: 10.1175/2007MWR2123.1.

W. C. Skamarock, J. B. Klemp, J. Dudhia, D. O. Gill, Z. Liu, J. Berner, W. Wang, J. G. Powers, M. G. Duda, D. M. Barker, and X.-Y. Huang. A description of the Advanced Research WRF model version 4. NCAR Technical Note NCAR/TN-556+STR, National Center for Atmospheric Research, Boulder, CO, 2019.

L. C. Slivinski, G. P. Compo, J. S. Whitaker, et al. Towards a more reliable historical reanalysis: Improvements for version 3 of the Twentieth Century Reanalysis system. *Quarterly Journal of the Royal Meteorological Society*, 145(724):2876–2908, 2019. doi: 10.1002/qj.3598.

L. C. Slivinski, G. P. Compo, P. D. Sardeshmukh, et al. An evaluation of the performance of the Twentieth Century Reanalysis version 3. *Journal of Climate*, 34(4):1417–1438, 2021. doi: 10.1175/JCLI-D-20-0505.1.

P. Stucki, P. Froidevaux, M. Zamuriano, F. A. Isotta, M. Messmer, and A. Martynov. Simulations of the 2005, 1910, and 1876 Vb cyclones over the Alps – sensitivity to model physics and cyclonic moisture flux. *Natural Hazards and Earth System Sciences*, 20(1):35–57, 2020. doi: 10.5194/nhess-20-35-2020.

P. Stucki, L. Pfister, Y. Brugnara, R. Varga, C. Hari, and S. Brönnimann. Dynamical downscaling and data assimilation for a cold-air outbreak in the European Alps during the Year Without a Summer of 1816. *Climate of the Past*, 20(10):2327–2348, 2024. doi: 10.5194/cp-20-2327-2024.

K. P. Sudheer, S. M. Bhallamudi, B. Narasimhan, J. Thomas, V. M. Bindhu, V. Vema, and C. Kurian. Role of dams on the floods of August 2018 in Periyar River Basin, Kerala. *Current Science*, 116 (5):780–794, 2019. doi: 10.18520/cs/v116/i5/780-794.

G. Thompson, P. R. Field, R. M. Rasmussen, and W. D. Hall. Explicit forecasts of winter precipitation using an improved bulk microphysics scheme. Part II: Implementation of a new snow parameterization. *Monthly Weather Review*, 136(12):5095–5115, 2008. doi: 10.1175/2008MWR2387.1.

M. Tiedtke. A comprehensive mass flux scheme for cumulus parameterization in large-scale models. *Monthly Weather Review*, 117(8):1779–1800, 1989. doi: 10.1175/1520-0493(1989)117<1779:ACMFSF>2.0.CO;2.

H. Wernli, M. Paulat, M. Hagen, and C. Frei. SAL—a novel quality measure for the verification of quantitative precipitation forecasts. *Monthly Weather Review*, 136(11):4470–4487, 2008. doi: 10.1175/2008MWR2415.1.

A All criteria outcomes

ID	Case	Values	Result
H1	WRF_3km_ERA5	FSS 0.55 (useful 0.60), bias −39%	fail
H1	RegCM_3km_ERA5	FSS 0.35 (useful 0.60), bias −33%	fail
H2	WRF_12km	dFSS 0.47, bias(20CRv3) −77%	fail
H2	WRF_3km	dFSS 0.55, bias(20CRv3) −84%	fail
H2	RegCM_12km	dFSS 0.34, bias(20CRv3) −79%	fail
H2	RegCM_3km	dFSS 0.31, bias(20CRv3) −58%	fail
H3	E1_kerala1924 RegCM_12km_20CRv3	AV +0.65	**pass**
H3	E1_kerala1924 RegCM_3km_20CRv3	AV +0.59	**pass**
H3	E1_kerala1924 WRF_12km_20CRv3	AV +0.71	**pass**
H3	E1_kerala1924 WRF_3km_20CRv3	AV +0.58	**pass**
H3	E2_mumbai2005 RegCM_12km_20CRv3	AV +0.05	**pass**
H3	E2_mumbai2005 RegCM_12km_ERA5	AV −0.02	fail
H3	E2_mumbai2005 RegCM_3km_20CRv3	AV +0.29	**pass**
H3	E2_mumbai2005 RegCM_3km_ERA5	AV −0.04	fail
H3	E2_mumbai2005 WRF_12km_20CRv3	AV +0.01	**pass**
H3	E2_mumbai2005 WRF_12km_ERA5	AV −0.03	fail
H3	E2_mumbai2005 WRF_3km_20CRv3	AV +0.03	**pass**
H3	E2_mumbai2005 WRF_3km_ERA5	AV −0.09	fail
H4	E1_kerala1924 WRF 20CRv3	peak 0.62→0.56, FSS 0.38→0.14	fail
H4	E1_kerala1924 RegCM 20CRv3	peak 0.52→0.62, FSS 0.17→0.25	**pass**
H4	E2_mumbai2005 WRF 20CRv3	peak 0.10→0.10, FSS 0.00→0.00	fail
H4	E2_mumbai2005 RegCM 20CRv3	peak 0.11→0.17, FSS 0.00→0.03	**pass**
H4	E2_mumbai2005 WRF ERA5	peak 0.46→0.54, FSS 0.47→0.55	**pass**
H4	E2_mumbai2005 RegCM ERA5	peak 0.23→0.41, FSS 0.34→0.35	**pass**
H5	WRF_3km_20CRv3	FSS 0.57 (useful 0.68), bias −53%	fail
H5	RegCM_3km_20CRv3	FSS 0.70 (useful 0.68), bias −39%	**pass**

B Event totals on the 3 km grids

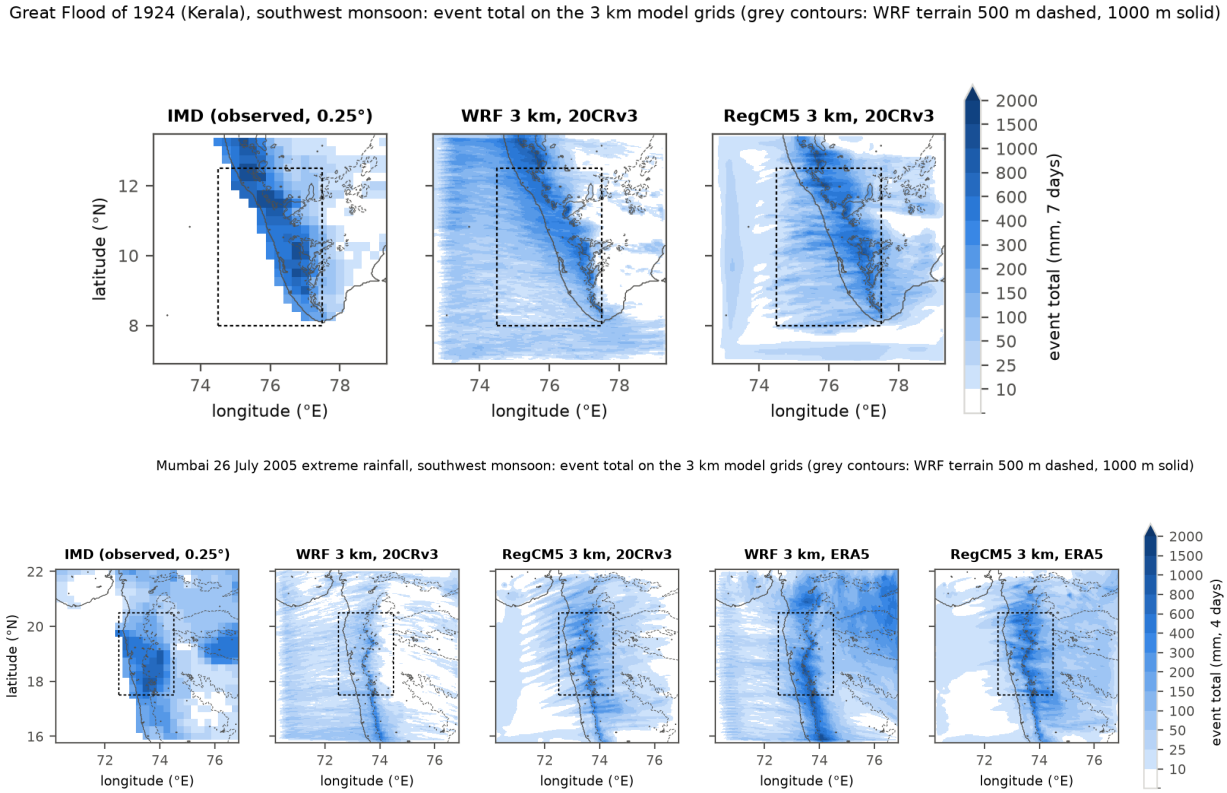


Figure 10. Event totals on the native 3 km grids with WRF terrain contours, E1 (top) and E2 (bottom). The east–west streaks over the sea lie along the low-level westerlies; the pale frame around the RegCM5 domain is its lateral relaxation zone. Neither affects the scores, which use IMD land cells only.